

Clinical Data Intelligence & Monitoring System

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Our team



Ziya Özkan

Clinical Research · 5+ years.
eCRF design, MedDRA, ICH
GCP, CDISC

Cosmina Scholz

Biostatistician · 18+ years in
CRO. Trial design, statistics,
SAS, SDTM & ADaM

Manish Kumar

Master's student in
Computer Engineering with
IoT systems

What we build



Clinical Data Intelligence & Monitoring

One platform for clinical prediction and document intelligence — built for researchers, data teams, and clinicians.



Patient Risk & Anomaly Checker

Patient-level cardiovascular risk with transparent, explainable models and record-level anomaly monitoring.

- CVD risk prediction (LightGBM)
- Per-patient explanation (SHAP)
- Anomaly monitoring (Isolation Forest)

ML

SHAP

Monitoring

[Open Patient Checker →](#)



Documents — CDIM System

Turn clinical study protocols into structured, validated data — with grounded Q&A and traffic-light review verdicts.

- Protocol parsing & RAG chat
- Extraction → Clean / Review / Block
- Risk predictions & insights

RAG

Extraction

Anti-hallucination

[Open Documents →](#)

Stack · LightGBM · SHAP · Isolation Forest · ChromaDB · Claude API · Streamlit

Trained on 68,595 cardiovascular records · ROC-AUC 0.8065

Patient Risk & Anomaly Checker

*Enter a patient's measurements
Get a risk score in under 1 second
See exactly which factors drove it
Flag unusual profiles automatically*



Documents CDIM System

Upload any clinical PDF
Ask questions in plain English
AI answers with page references



Live demo



Thanks!

Q&A session – do you have any questions?



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Cosmina Scholz

Manish Kumar



The Data – 70,000 Real Patient Records

Cardiovascular dataset - Kaggle

70,000
Patient records

68,595
After cleaning

49,9%
Have CVD

12
Features used

What we measured about each patient

- Age – 20 - 90 years
- Blood Pressure – systolic + diastolic
- Glucose level – normal / above / well above
- Alcohol intake – yes / no
- Height & Weight – used to compute BMI
- Cholesterol level – normal / above / well above
- Smoking – yes / no
- Physical activity – yes / no

Model Results – Cardiovascular dataset (68K patients, 20% test set)

Classifier	ROC-AUC	F1	Accuracy
LightGBM ★	0.8065	0.7198	0.7354
XGBoost	0.7961	0.7167	0.7336
Logistic Regression	0.7950	0.7092	0.7304
Random Forest	0.7759	0.7088	0.7167